

What does it cost to train a small language model?

Training speed and price on rented and own GPUs

Arkadiusz Słota*

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Abstract

We want to answer one practical question: to train a model of a given size on a given number of tokens, which GPU should we rent, how long will it take and what will it cost? This first version uses only measurements from our own past training runs (models from 31 M to 253 M parameters, on RTX 5090, RTX Pro 6000 and Radeon RX 7900 XTX) and public RunPod prices from 2 October 2026. Three findings: for models up to about 64 M parameters the speed is limited by the CPU of the host machine, not by the GPU, so the same RTX 5090 gives anywhere from 250 000 to 363 000 tokens per second; the RTX 5090 is the cheapest GPU per token in everything we have measured so far (0.53–2.14 USD per billion tokens); two GPUs train faster than one but cost 25 % more per token. The next version adds GPUs we have not used yet (RTX 3090, 4090, A100, H100 and others).

1 The question

A small language model has between 30 and 500 million parameters (N). It is trained on some number of tokens; a token is a piece of text, on average about four bytes of Polish or English. We give the number of tokens as a multiple of N : $20N$ is the usual minimum for a model of this size, $100N$ is a long training run; our own final runs used 200–400 N . For each model size we want three numbers: tokens per second on a given GPU, hours to finish, and the price.

2 How we measured

Speed. We measure training throughput in tokens per second: the median over the run, or over a short benchmark when no full run exists, leaving out the moments when the model is being evaluated or saved. All numbers come from our own training code: bf16 precision, `torch.compile`, context of 1 024 tokens, batch of 32 sequences. Other code will give other numbers.

Price. We use the public RunPod price list as of 2 October 2026, 12:38 UTC. RunPod has two kinds of machines: *community* (hosted by private owners, cheaper) and *secure* (data centres, more expensive). The cost of training is

$$\text{USD per billion tokens} = \frac{\text{USD per hour}}{\text{tokens per second} \times 3600} \times 10^9.$$

It covers training time only: setting up the machine, evaluating the model and idle time are not included, so a real bill is higher. Prices change every few weeks; the formula lets anyone recompute the tables with new prices.

3 Results

Table 1 gives speed, cost per billion tokens, and hours and cost to reach $20N$ and $100N$ tokens for every combination of model and GPU we have trained on.

*Prepared with the Kolektyw AI agents: Hart (measurements and numbers), Latarnik (draft), Wartownik (prices and review), Monter (training infrastructure).

model	GPU (host CPU)	tokens/s	USD per 1 B tokens	20N: hours / USD	100N: hours / USD
31 M	RTX 5090 (Ryzen 9950X)	363 000	0.53	0.5 / 0.3	2.4 / 1.7
31 M	RTX 5090 (EPYC 7543)	304 000	0.63	0.6 / 0.4	2.9 / 2.0
31 M	RTX 5090 (Xeon 6530)	250 000	0.77	0.7 / 0.5	3.5 / 2.4
31 M	RTX Pro 6000	337 000	1.39	0.5 / 0.9	2.6 / 4.4
63 M	RTX 5090	222 000	0.86	1.6 / 1.1	7.9 / 5.4
123 M	RTX 5090	140 500	1.36	4.9 / 3.4	24.3 / 16.8
248 M	RTX 5090 (older code)	89 400	2.14	15.4 / 10.6	77 / 53
253 M	RTX Pro 6000	99 100	4.74	14.2 / 23.9	71 / 120
253 M	2 × RTX Pro 6000	158 600	5.92	8.9 / 29.9	44 / 150

Table 1: Training speed and cost at community prices (RTX 5090 0.69 USD/h, RTX Pro 6000 1.69 USD/h per GPU). Secure prices are 43 % higher for the RTX 5090 and 24 % higher for the RTX Pro 6000.

We also trained a 110 M model on our own Radeon RX 7900 XTX (24 GB, ROCm under WSL2), with an older training script and a context of 512 tokens: 66 500–72 800 tokens per second, about half of what the RTX 5090 gives for a model of similar size. This GPU has no hourly price; it costs only electricity.

4 What follows from the numbers

1. **Up to about 64 M parameters, the host CPU sets the speed.** The same RTX 5090 trains our 31 M model at 363 000, 304 000 or 250 000 tokens per second depending on the processor of the machine it sits in. Our training loop keeps one CPU core fully busy, and for a small model the GPU waits for it. A more expensive GPU does not help: the RTX Pro 6000 is as fast as the RTX 5090 on the 31 M model, costs 2.45 times more per hour, and 1.8–2.6 times more per token.
2. **Larger models use the GPU better.** On the RTX 5090 the work done per second grows from about 84 to 104 and 133 trillion operations per second for the 63 M, 123 M and 248 M models (estimated from the number of parameters, so a lower bound).
3. **Two GPUs are faster, not cheaper.** Two RTX Pro 6000 cards train the 253 M model 1.6 times faster than one, and each token costs 25 % more.
4. **Compilation matters.** Without `torch.compile` the 253 M model trains 1.65 times slower, so the same work costs 1.65 times more.
5. **The RTX 5090 is the cheapest per token in everything we measured,** from 31 M to 248 M parameters. For the 248–253 M models the comparison is not yet clean: the RTX 5090 number comes from an older version of our code.

5 What we got for the money

Our own finished runs, scored with our evaluation protocol (*our protocol, not official*; “eff” is the average of our English benchmarks). Cost is training time at the community prices of Section 2.

run	tokens	hours	cost (USD)	eff
63 M, shorter run	13.1 B	17.8	12.3	75.81
63 M, longer run	24.9 B	33.6	23.2	76.07
123 M, shorter run	13.1 B	26.6	18.3	75.84
123 M, longer run	24.9 B	49.2	34.0	76.94

For the 63 M model the longer run gave +0.26 points for 10.9 USD more; for the 123 M model it gave +1.10 points for 15.7 USD more. The larger model makes better use of extra data. The shorter and longer runs also differ in recipe, so this is a cost observation, not a controlled experiment. A 253 M model is still training (19.7 B tokens and about 93 USD so far); its numbers are *training in progress; not a result*.

6 What we do not know yet

- We have no measurements on RTX 3090, RTX 4090, RTX A6000, L40S, A100 or H100, and none with eight GPUs. The next version adds them, measured with one benchmark script on synthetic data.
- The 248 M model on the RTX 5090 was measured with an older version of our code; the next version repeats it.
- The same GPU model varies by about $\pm 20\%$ between host machines (rule 1). One number per GPU is not enough; we will report the host CPU next to every measurement.
- A container on a rented machine can see more CPUs and memory than it may use: on one RTX 5090 machine the system tools showed 256 CPUs and 503 GB of memory, while the container was limited to 35 CPUs and 71 GB. Plans based on the larger numbers fail.

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